

Green chemistry

Introduction:

"It deals with the environment friendly compounds."

- * Removal of hazardous substances."

Definition:

"The design of chemical processes which reduces or eliminate the use or generation of hazardous substances is called green chemistry."

Attributes of green chemistry:

- * simplicity
- * High Yield
- * use of renewable or recyclable materials.
- * safety or safe to use.
- * Absence of hazardous

Principles of green

chemistry:

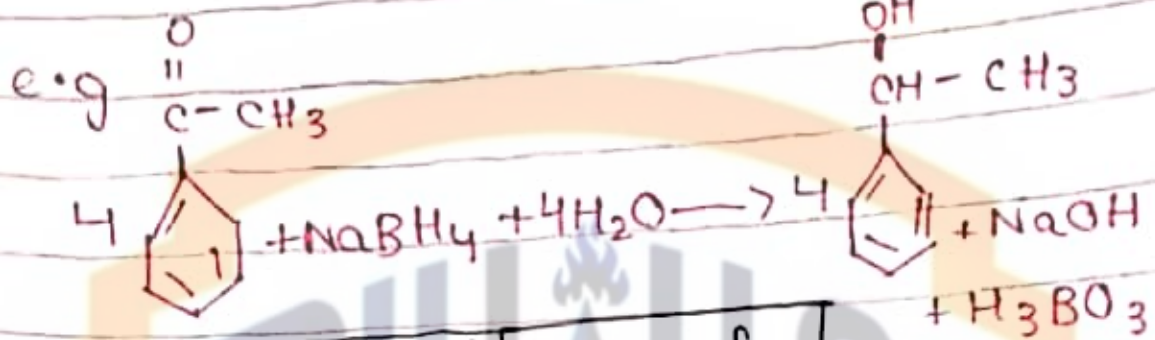
100% Environment friendly
less hazardous product:

- * Waste minimization
- * Safe chemical processes
- * nontoxic materials
- * Environmentally friendly
- * use of less hazardous material.
- * Prevention
- * Real time analysis.
- * increase or reduction of chemical derivation.
- * Design biodegradable processes.
- * use of safe chemicals.
- * use of catalyst
- * Technology.

Atom Economy:

It deals with
the conversion of large
numbers of reactants
into products.

$$\% \text{ atom economy} = \frac{\text{F.W of final products}}{\text{F.W of Reactants}} \times 100\%$$

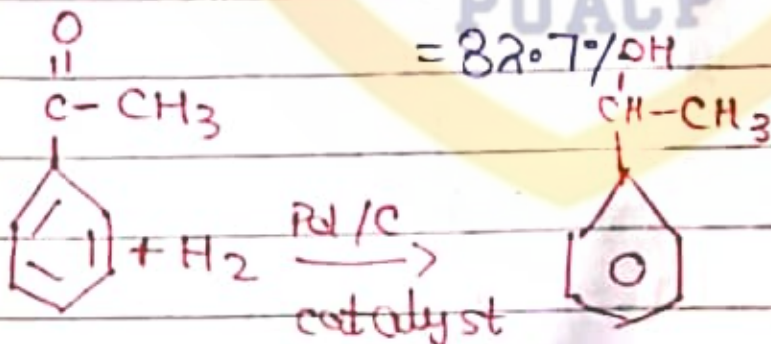


M.F of reagent	M.W of reagent	F.W of product
C ₈ H ₈ O	4 × 120 = 480	C ₈ H ₁₀ O = 488
NaBH ₄	38	
H ₂ O	4 × 18 = 72	

590

$$\% \text{ atom economy} = \frac{488}{590} \times 100\%$$

= 82.7%



C ₈ H ₈ O	120	C ₈ H ₁₀ O = 122
H ₂	2	
	122	

$$\% \text{ economy} = \frac{122}{122} \times 100\% \\ = 100\%$$

Improved process:

green chemistry
emphasize the need to use
raw materials that are
renewable.

e.g:

For electricity generation
instead of green chemist
suggest agricultural waste
like bagasse, rice husk etc.

Minimize toxicity:

green chemistry
aims at adopting such
products which use &
produce those substances
which are not toxic or
hazardous and do not
(toxic) harm living
organism and
environment.

Environmental Friendly or use of safe solvents:

volatile or toxic
solvents are health hazards.
For example:

CCl_4 & benzene
are used as solvents but
they are carcinogenic in
nature. so instead of
these solvents like water
or liquid CO_2 .

less Hazardous chemical Synthesis

chemicals should be
design in such a way that
they do not cause severe
accident like explosion &
form non-toxic products
in case of accident.

Prevention:

The main
principles of green

chemistry is preventing or reducing the waste generation. This is because sometimes disposal of waste becomes so expensive & associated with problems like waste storage & transportation.

e.g: use of fossil fuels to get energy generate green house gases (CO_2 & NH_4)

Now, instead of heating CO_2 after production, green chemistry need to develop a technique that can reduce CO_2 production.

This can be done by the use of renewable energy resources like solar energy or wind energy.

Real time Analysis:

Analytical method should be adopted for

monitoring or reducing the formation of toxic substance.
Omit or reduce chemical derivatives:

Green chemistry omits or reduces the extra steps as extra steps requires extra reagents which generate extra wastes.

Design biodegradable products:

chemical substances should be designed in such away that they degrade in the environment to give harmless products.

e.g:

- 1) Designing biodegradable plastics instead of plastic.
- 2) use of Diacyl hydrazine (Renewable) instead of DDT (Non-renewable insecticide).

USE of safe chemicals:

The reaction should be designed in such a way that they should be fully safe and effective without any toxic effect.

e.g.

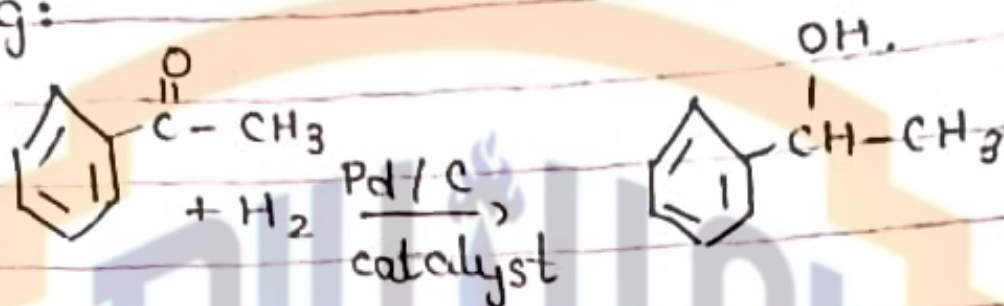
chlorofluorocarbons cause ozone depletion due to the generation of free radicals. To prevent this chemists have designed hydrofluorocarbons in which chlorine atoms are replaced by hydrogen atoms. The ozone depletion potential of this compound is far less than CFC.

Catalyst USE:

use of selective

catalyst increases the specificity of the reaction & reduces the formation by bi-product.

e.g.:



Technology:

There are many chemical reactions that requires a large amount of energy like preparation of cement and glass require high temperature ($1600-1800^\circ\text{C}$)

This energy generally comes from the use of fossil fuels.

→ green chemistry emphasize the development of

Such chemical reactions that can be carried out at room temperature & pressure, However energy to a reaction can be supplied by photochemical means, microwaves & photocells- etc.

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